

MODULE FOUR

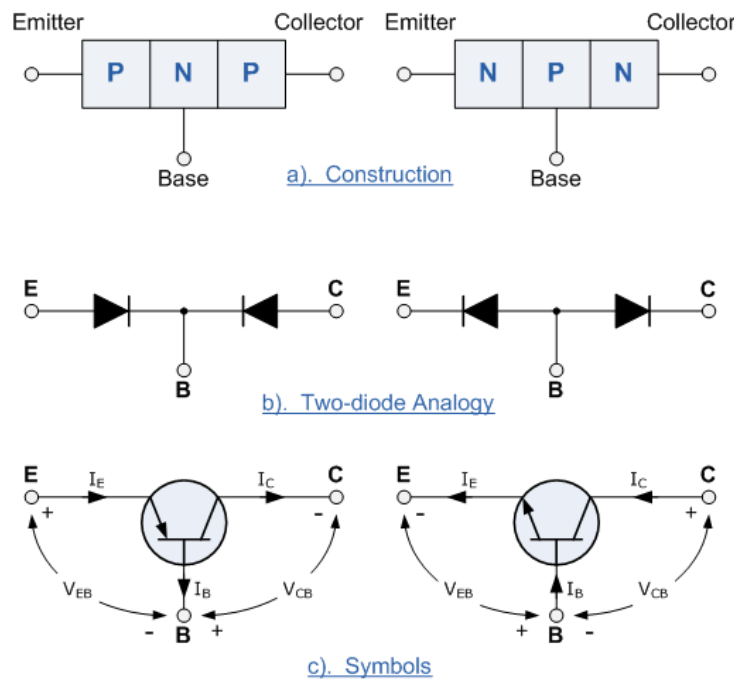
BIPOLAR JUNCTION TRANSISTORS

A transistor is a sandwich of one type of semiconductor (P-type or n-type) between two layers of other types.

Transistors are classified into two types;

1. **PNP** transistor: It is obtained when a n-type layer of silicon is sandwiched between two p-type silicon material.
2. **NPN** transistor: it is obtained when a p-type layer of silicon is sandwiched between two n-type silicon materials.

Figure3.1 below shows the schematic representations of a transistor which is equivalent of two diodes connected back to back.



The three portions of transistors are named as emitter, base and collector. The junction between emitter and base is called emitter-base junction while the junction between the collector and base is called collector-base junction.

The base is thin and tightly doped, the emitter is heavily doped and it is wider when compared to base, the width of the collector is more when compared to both base and emitter.

In order to distinguish the emitter and collector an arrow is included in the emitter. The direction of the arrow depends on the conventional flow of current when emitter base junction is forward biased.

In a *pnp* transistor when the emitter junction is forward biased the flow of current is from emitter to base hence, the arrow in the emitter of *pnp* points towards the base.

3.1 Operating regions of a transistor

A transistor can be operated in three different regions as

- a. active region
- b. saturation region
- c. cut-off region

Active region

The transistor is said to be operated in active region when the emitter-base junction is forward biased and collector base junction is reverse biased. The collector current is said to have two current components one is due to the forward biasing of *EB* junction and the other is due to reverse biasing of *CB* junction. The collector current component due to the reverse biasing of the collector junction is called reverse saturation current and it is very small in magnitude.

Saturation region

Transistor is said to be operating in saturation region when both *EB* junction and *CB* junction are forward biased as shown. When transistor is operated in saturation region I_C increases rapidly for a very small change in V_C .

Cut-off region

When both *EB* junction and *CB* junction are reverse biased, the transistor is said to be operated in cut-off region. In this region, the current in the transistor is very small and thus when a transistor in this region it is assumed to be in off state.

3.2 Working of a transistor (*NPN*)

Consider a *nnp* transistor operated in active region as shown in Figure 3.6

Since the *EB* junction is forward biased large number of holes present in the emitter as majority carriers are repelled by the +ve potential of the supply voltage V_{EB} and they move towards the base region causing emitter current I_E .

Since the base is thin and lightly doped very few of the holes coming from the emitter recombine with the electrons causing base current I_B and all the remaining holes move towards the collector. Since the *CB* junction is reverse biased all the holes are immediately attracted by the positive potential of the supply V_{CB} . Thereby giving rise to collector current I_C .

Thus $I_E = I_B + I_C$

Since the *CB* junction is reversed biased a small minority carrier current I_{co} flows from the base to collector.

Fig 3.7 above shows a transistor operated in active region. It can be noted from the diagram the battery V_{EB} forward biases the EB junction while the battery V_{CB} reverse biases the CB junction. As the EB junction is forward biased the holes from emitter region flow towards the base causing a hole current I_{PE} . At the same time, the electrons from base region flow towards the emitter causing an electron current I_{NE} . Sum of these two currents constitute an emitter current $I_E = I_{PE} + I_{NE}$.

The ratio of hole current I_{PE} to electron current I_{NE} is directly proportional to the ratio of the conductivity of the p-type material to that of n-type material. Since, emitter is highly doped when compared to base; the emitter current consists almost entirely of holes.

Not all the holes, crossing EB junction reach the CB junction because some of them combine with the electrons in the n-type base. If I_{PC} is the hole current at (Jc) CB junction. There will be a recombination current $I_{PE} - I_{PC}$ leaving the base as shown in figure 3.7.

If emitter is open circuited, no charge carriers are injected from emitter into the base and hence emitter current $I_E = 0$. Under this condition CB junction acts as a reverse biased diode and therefore the collector current ($I_C = I_{CO}$) will be equal to the reverse saturation current. Therefore when EB junction is forward biased and collector base junction is reverse biased the total collector current $I_C = I_{PC} + I_{CO}$.

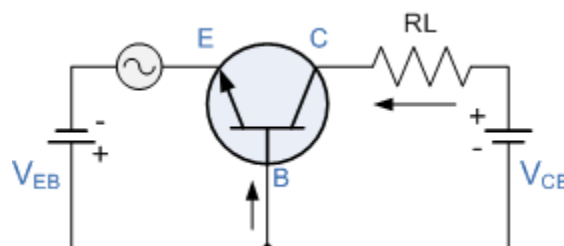
Transistor Configuration

There are three possible ways of connecting bipolar transistor in an electronic circuit with each method of connection responding differently to its input signal as the static characteristics of the transistor vary with each circuit arrangement.

- 1. Common Base Configuration - has Voltage Gain but no Current Gain.
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- 2. Common Emitter Configuration - has both Current and Voltage Gain.
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- 3. Common Collector Configuration - has Current Gain but no Voltage Gain.

1. Common Base Configuration

A simple circuit arrangement of CB configuration for pnp transistor is shown below.



In this configuration, base is used as common to both input and output. It can be noted that the input section has an *a.c.* source V_i along with the *d.c.* source V_{EB} . The purpose of including V_{EB} is

to keep EB junction always forward biased (because if there is no V_{EB} then the EB junction is forward biased only during the positive half-cycle of the input and reverse biased during the negative half cycle. In CB configuration, I_E is the input current, I_C is the output current.

Current Relations

1. Current amplification factor (α)

It is defined as the ratio of d.c. collector current to d.c. emitter current

$$\alpha = \frac{I_C}{I_E}$$

2. Total input-output (o/p) current

We know that CB junction is reverse biased and because of minority charge carriers a small reverse saturation current I_C flows from base to collector.

$$I_E = I_B + I_C$$

$$I_C = \alpha I_E$$

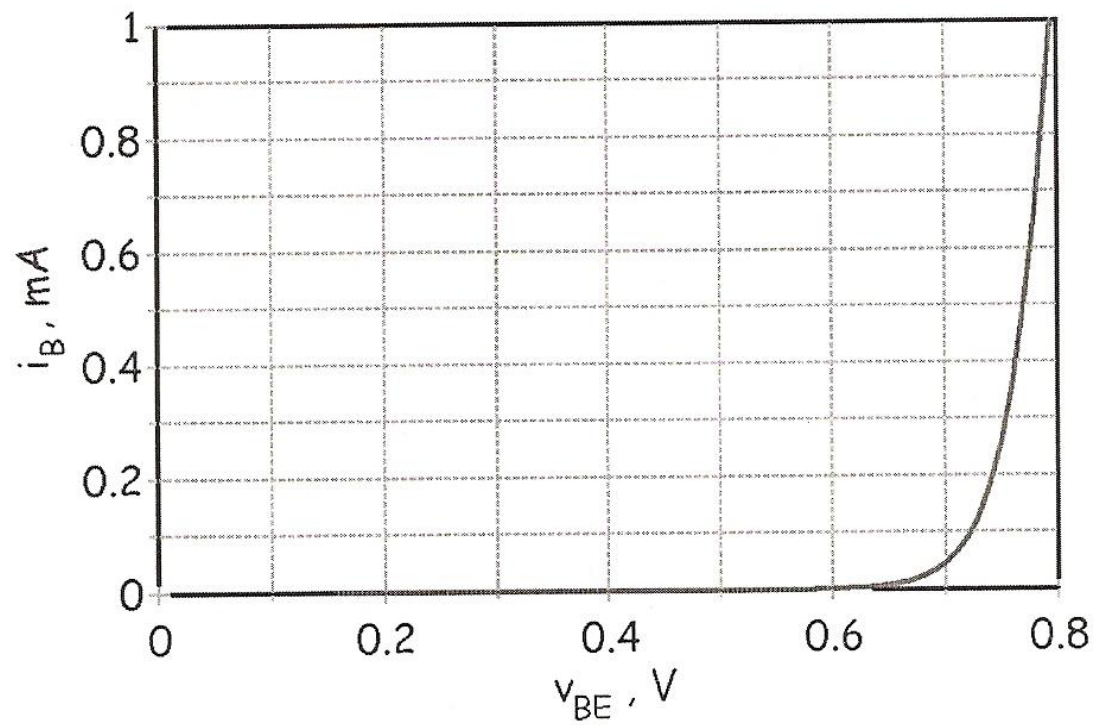
$$I_E = \alpha I_E + I_C$$

Transistor Characteristics

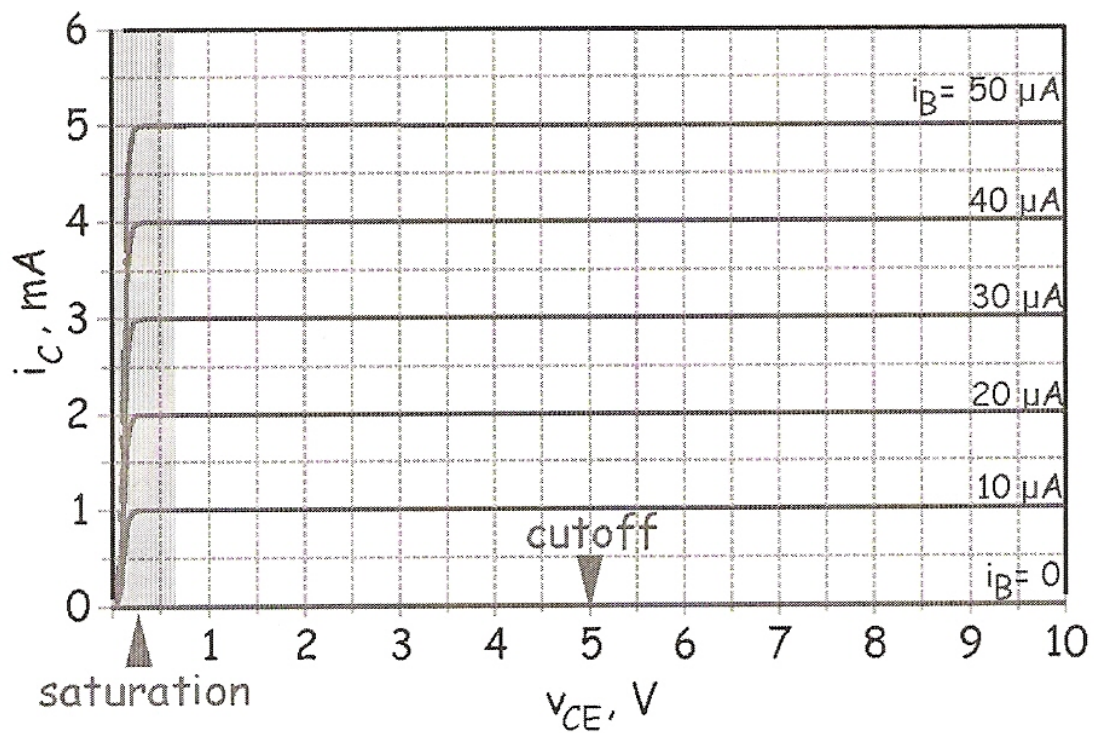
1. Input characteristics

The input characteristic is a curve between I_E and emitter base voltage V_{EB} keeping V_{CB} constant. I_E is taken along y-axis and V_{EB} is taken along x-axis. From the graph following points can be noted.

1. For small changes of V_{EB} there will be a large change in I_E . Therefore input resistance is very small.
2. I_E is almost independent of V_{CB} .
3. Input resistance, $R_i = \frac{V_{EB}}{I_E}$, with $V_{CB} = \text{constant}$



2. Output characteristics



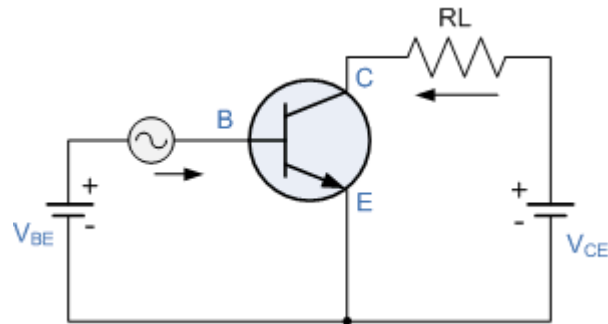
Output characteristics is the curve between I_C and V_{CE} at constant I_B . The collector current I_C is taken along y-axis and V_{CE} is taken along x-axis. It is clear from the graph that the output current I_C remains almost constant even when the voltage V_{CE} is increased.

i.e, a very large change in V_{CB} produces a small change in I_C . Therefore, output resistance is very high.

Output resistance $R_o = \frac{V_{EB}}{I_C}$, with $I_E = \text{constant}$

Region below the curve $I_E = 0$ is known as cut-off region where I_C is nearly zero. The region to the left of $V_{CB} = 0$ is known as saturation region and to the right of $V_{CB} = 0$ is known as active region.

2. CE configuration



In this configuration the input is connected between the base and emitter while the output is taken between collector and emitter. For this configuration I_B is input current and I_C is the output current. The common emitter is an inverting amplifier circuit resulting in the output signal being 180° out of phase with the input voltage signal.

1. Current amplification factor (β)

It is the ratio of *d.c.* collector current to *d.c.* base current.

$$\text{i.e., } \beta = \frac{I_C}{I_B}$$

2. Relationship between α and β

$$\text{We know } \alpha = \frac{I_C}{I_E}$$

$$\alpha = \frac{I_C}{I_E + I_C}$$

Divide both numerator and denominator of *RHS* by I_C , we get.

$$\alpha = \frac{1}{\frac{I_E}{I_C} + 1}$$

$$\alpha = \frac{1}{\frac{1}{\beta} + 1}$$

$$\beta = \frac{I_C}{I_B}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

Also

$$\alpha(1 + \beta) = \beta$$

$$\alpha + \alpha\beta = \beta$$

$$\alpha = \beta(1 - \alpha)$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

Transistor characteristic

Input characteristic curve

Input characteristics is a curve between V_{BE} and base current I_B at constant V_{CE} . From the graph following can be noted.

1. The input characteristic resembles the forward characteristics of a p-n junction diode.
2. For small changes of V_{BE} there will be a large change in base current I_B . i.e., input resistance is very small.
3. The base current is almost independent of V_{CE} .
4. Input resistance, $R_i = \frac{V_{BE}}{I_B}$ with $V_{CE} = \text{constant}$

It is the curve between V_{CE} and I_C at constant I_B . From the graph we can see that,

1. Very large changes of V_{CE} produces a small change in I_C i.e. output resistance is very high.
2. Output resistance $R_o = \frac{V_{CE}}{I_C}$ with $I_B = \text{constant}$

Region between the curve $I_B = 0$ is called cut-off region where I_B is nearly zero. Similarly the active region and saturation region is shown on the graph.

Common Collector Configuration.

In this configuration the input is connected between the base and collector while the output is taken between emitter and collector. Here I_B is the input current and I_E is the output current.

Current relations

1. Current amplification factor (γ)

2. Relationship between α , β and γ

$$\gamma = \frac{I_E}{I_B}$$

$$\gamma = \frac{I_B + I_C}{I_B}$$

Dividing both Numerator and denominator by I_B

$$\gamma = \frac{1 + \frac{I_C}{I_B}}{1}$$

$$\gamma = 1 + \beta \quad \left(\beta = \frac{I_C}{I_B} = \frac{\alpha}{1 - \alpha} \right)$$

$$\gamma = 1 + \frac{\alpha}{1 - \alpha}$$

Comparison of CB, CE, CC

Characteristics	Common Base	Common Emitter	Common Collector
Input resistance (R_i)	Low	Low	High
Output resistance (R_o)	High	High	Low
Current amplification factor	$\alpha = \frac{\beta}{1 + \beta}$	$\beta = \frac{\alpha}{1 - \alpha}$	$\gamma = \frac{\alpha}{1 - \alpha}$
Phase relationship between input and output	In phase	Out of phase	In phase
Application	High frequency applications	Audio frequency application	Impedance matching
Current gain	Less than Unity	Greater than unity	Very high
Voltage gain	High	Greater than unity	Less than unity

Worked Example

1. An NPN Transistor has a DC current gain, $\beta = 200$. Calculate the base current I_B required switch a resistive load of 4 mA .

$$\beta = \frac{I_C}{I_B}$$

$$I_B = \frac{I_C}{\beta} = \frac{4 \times 10^{-3}}{200} = 20\mu\text{A}$$

Therefore, $\beta = 200$, $I_C = 4\text{mA}$ and $I_B = 20\mu\text{A}$.

Note: The collector voltage, V_C must be greater than the emitter voltage, V_E to allow current to flow through the device between the collector-emitter junction. Then the base voltage, V_{BE} of an Transistor must be greater than this 0.7 V otherwise the transistor will not conduct with the base current given as.

$$I_B = \frac{V_B - V_{BE}}{R_B}$$

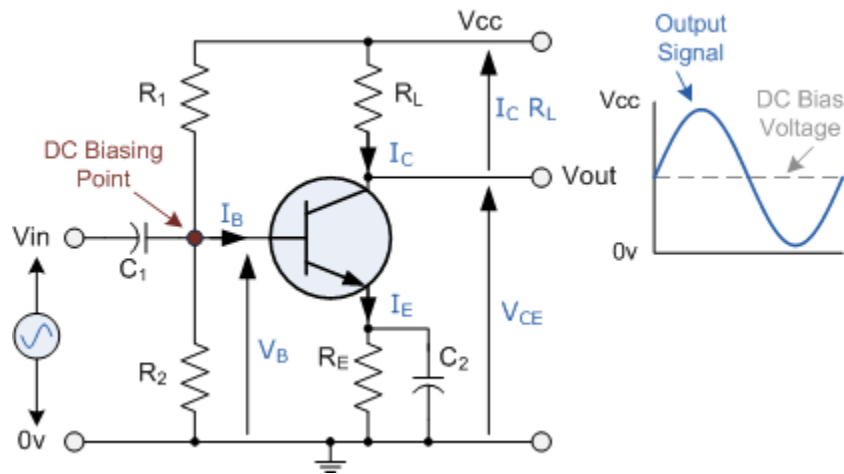
Worked Example.

2. An NPN Transistor has a DC base bias voltage, $V_B = 10\text{V}$ and an input base resistor, R_B of $100\text{k}\Omega$. What will be the value of the base current into the transistor?

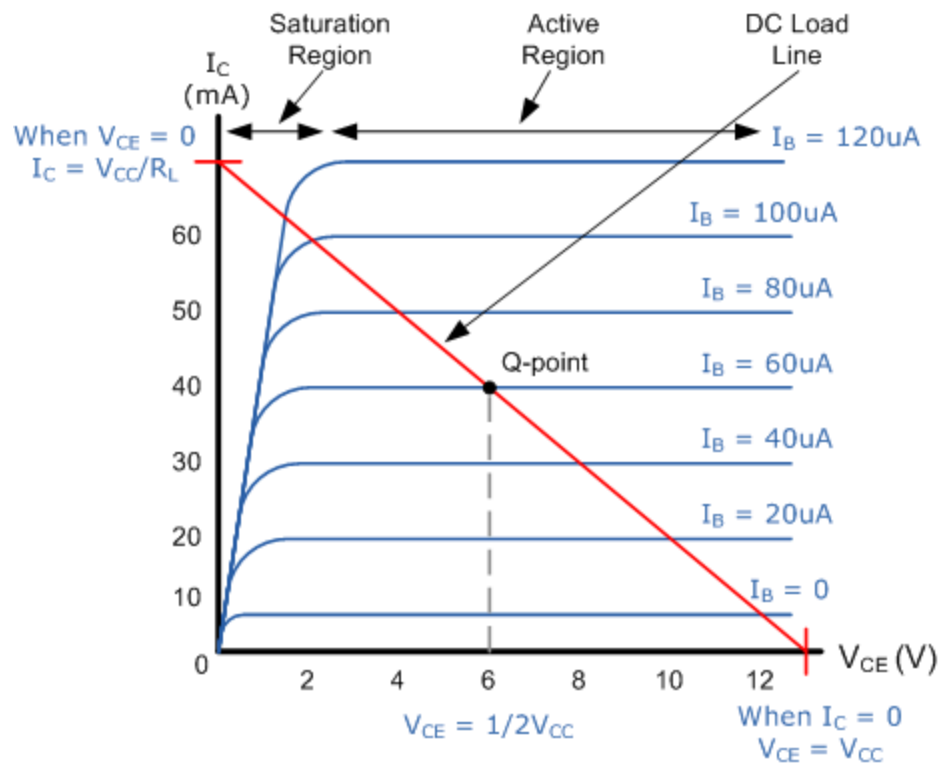
$$I_B = \frac{V_B - V_{BE}}{R_B} = \frac{10 - 0.7}{100 \times 10^3} = 93\mu\text{A}$$

The AC-DC Load Lines.

Using the common emitter configuration shown below, a family of curves known commonly as the Output Characteristics Curves, relates the output collector current, (I_C) to the collector voltage, (V_{ce}) when different values of base current, (I_b) are applied to the transistor for transistors with the same β value. A DC Load Line can also be drawn onto the output characteristics curves to show all the possible operating points when different values of base current are applied. Initial V_{ce} is correctly set to allow the output voltage to vary when amplifying AC input signals. This is normally referred to as setting the operating point or Quiescent Point (Q-point).



Output Characteristics Curves for a Typical Bipolar Transistor



The effect of V_{ce} on the collector current I_c when V_{ce} is greater than about 1.0 volts is that I_c is largely unaffected by changes in V_{ce} above this value and instead it is almost entirely controlled by the base current, I_b .

It can also be seen from the emitter current I_e is the sum of the collector current, I_c and the base current, I_b , added together so we can also say that $I_e = I_c + I_b$.

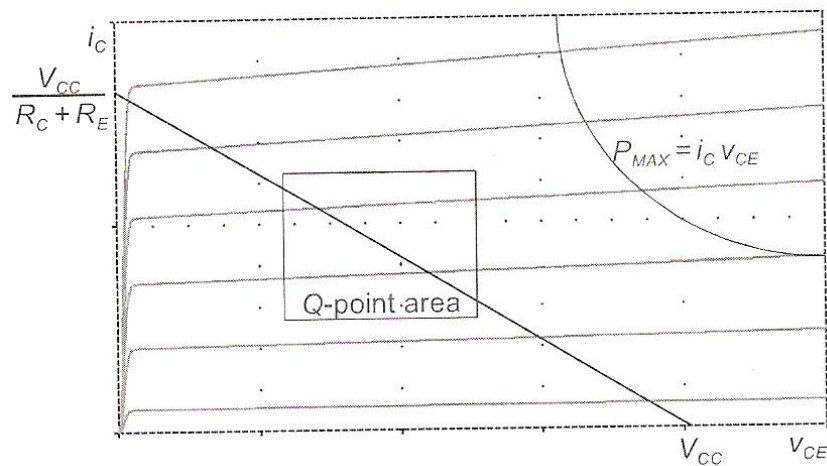
By using the output characteristics curves and Ohm's Law, the current flowing through the load resistor, (R_L), is equal to the collector current, I_c entering the transistor which in turn corresponds to the supply voltage, (V_{cc}) minus the voltage drop between the collector and the emitter terminals, (V_{ce}) and is given as:

$$I_C = \frac{V_{CC} - V_{CE}}{R_L}$$

Also, a Load Line can be drawn directly onto the graph of curves above from the point of saturation when $V_{CE} = 0$ to the point of Cut-off when $I_C = 0$ giving us the Operating or Q-point of the transistor. These two points are calculated as:

$$\text{When } V_{CE} = 0, I_C = \frac{V_{CC} - 0}{R_L} = \frac{V_{CC}}{R_L}$$

$$\text{When } I_C = 0, 0 = \frac{V_{CC} - V_{CE}}{R_L} \text{ and } V_{CC} = V_{CE}$$



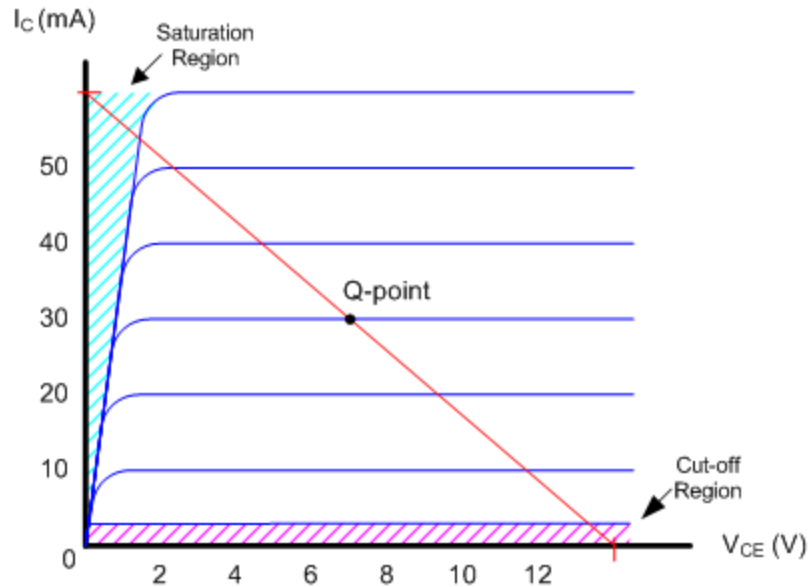
Identifying Transistor using Resistance Values.

The table below gives the summary of the simplest way to identify resistors and their terminals.

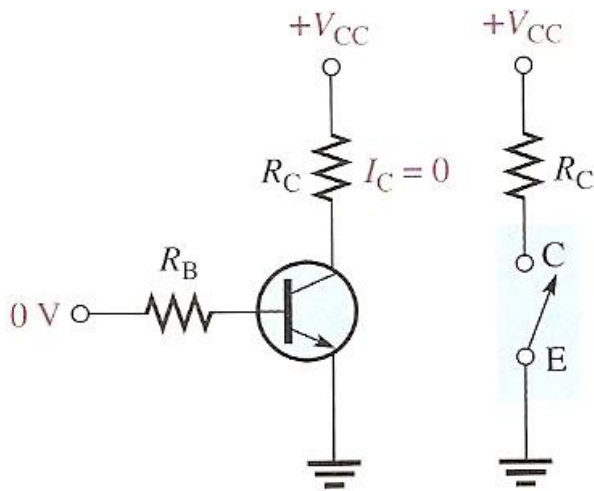
Between Transistor Terminals		PNP	NPN
Collector	Emitter	R_{HIGH}	R_{HIGH}
Collector	Base	R_{LOW}	R_{HIGH}
Emitter	Collector	R_{HIGH}	R_{HIGH}
Emitter	Base	R_{LOW}	R_{HIGH}
Base	Collector	R_{HIGH}	R_{LOW}
Base	Emitter	R_{HIGH}	R_{LOW}

The Transistor as a Switch

BJT can be made to operate as a switch. If the circuit uses the Transistor as a Switch, then the biasing is arranged to operate in Saturation (ON) and Cut-off (OFF).



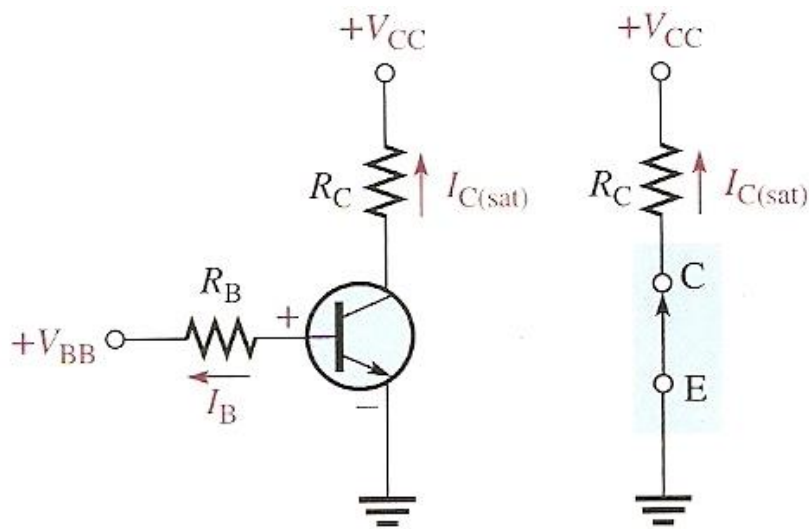
Cut-off Region - Both junctions are Reverse-biased, Base current is zero or very small resulting in zero Collector current flowing, and the device is switched fully OFF.



V_{CE} is approximately equal to V_{CC}

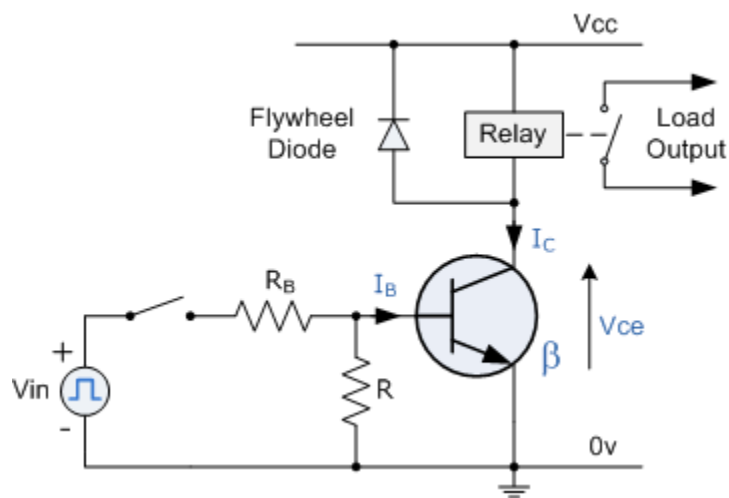
$$V_{CE}(\text{cut-off}) = V_{CC}$$

Saturation Region - Both junctions are Forward-biased, Base current is high enough to give a Collector-Emitter voltage of 0V resulting in maximum Collector current flowing, and the device is switched fully ON.



$$I_{C(saturation)} = \frac{V_{CC}}{R_C}$$

An example of an NPN Transistor as a switch being used to operate a relay is given below. The diode to dissipate the back EMF generated by the inductive load when the transistor switches OFF and also protects the transistor from damage.



Example No1.

For example, using the transistor values from the previous tutorials of: $\beta = 200$, $I_c = 4 \text{ mA}$ and $I_b = 20 \mu\text{A}$, find the value of the Base resistor (R_b) required to switch the load ON when the input terminal voltage exceeds 2.5 V .

$$R_b = \frac{V_{in} - V_{be}}{I_b} = \frac{2.5 - 0.7}{20 \times 10^{-6}} = 90 \text{ k}\Omega$$

Example No2.

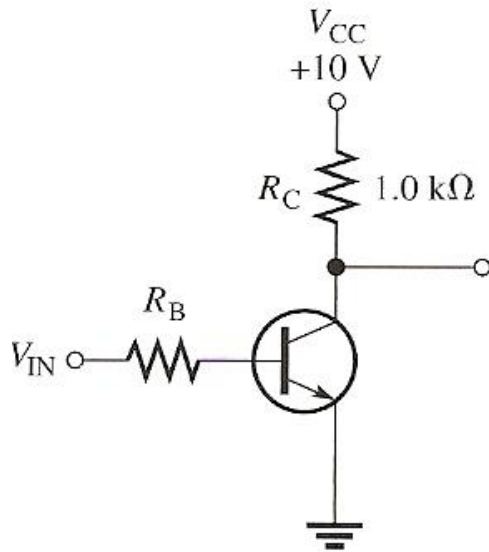
Again using the same values, find the minimum Base current required to turn the transistor fully ON (Saturated) for a load that requires 200 mA of current.

$$I_b = \frac{I_c}{\beta} = \frac{200 \times 10^{-3} A}{200} = 1mA$$

Example 3

Use the circuit below to answer the questions that follows.

- For the transistor switch circuit below what is V_{CE} , $V_{in} = 0V$
- What is the minimum value of I_B will saturate of this transistor if the $\beta = 200$.
- Calculate the maximum value of R_B when $V_{in} = 5V$

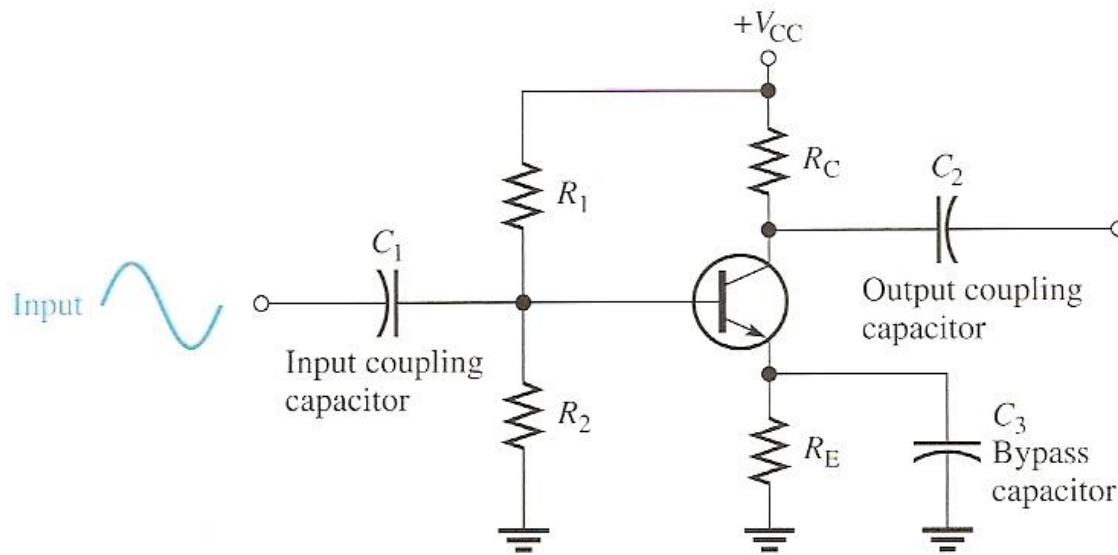


Solution:

Transistor as an Amplifier

In order to use a transistor as an amplifier it should be operated in active region i.e. emitter junction should be always forward bias and collector junction should be reversed bias. Therefore in addition to the *a.c.* input source V_i , two *d.c.* voltages V_{BE} and V_{CE} are applied as shown. This *d.c.* voltage is called bias voltage.

As the input circuit has low resistance, a small change in the input signal causes a large change in the base current thereby causing the same change in collector current. Thus a weak signal applied at the input circuit appears in the amplified form at the output.



C_1 and C_2 are coupling capacitors used to pass the signal into and out of the amplifier such that it will not affect the bias voltages. C_3 is a bypass capacitor that shorts the emitter signal voltage (ac) to ground without disturbing the emitter voltage. Because of the bypass of capacitor, the emitter is at signal (ac) ground (but not dc ground), thus making the circuit a common emitter amplifier. The bypass capacitor increases the signal voltage gain.

A standard DC analysis of four-resistor bias circuit provides the Q-point, and from that we obtain the value load lines.

